

Appendix A Terminology and RL Reproducibility

This appendix provides terminology definitions and full RL implementation details referenced in Section 3.

A.1 Terminology Definitions

Table A1 defines the terms used consistently throughout the manuscript. To prevent ambiguity, we use the following terminology consistently throughout the paper.

Table A1 Terminology used in this study.

Term	Definition in this paper
Training episode	One complete computational simulation run from initialization to attacker win or tick cutoff ($T_{\max} = 1000$).
Training duration	Number of RL training episodes completed (e.g., 100, 500, 1,000, 2,000, 5,000).
RL training	Computational policy optimization of Rainbow DQN parameters across episodes.
Human-team training (analogy)	Conceptual interpretation of RL learning phases as a proxy for human team development; this is an analogy, not a calibrated real-time equivalence.
Baseline / heuristic / untrained	Same condition in this paper: fixed hand-crafted rules with no parameter updates and no cross-episode learning from experience.
Checkpoint evaluation	Standardized offline evaluation of frozen policies on the fixed benchmark grid with learning disabled ($\varepsilon = 0$).
Training-log curve	On-policy learning signal computed from training episodes under curriculum and exploration settings.

A.2 Rainbow DQN Architecture

Table A2 lists the attacker/scout network architecture used in all reported runs.

Table A2 Neural network architecture used by both Rainbow agents.

Component	Structure	Notes
Input state size	Attacker: $13 + 9N + 14$; Scout: $13 + 9N + 16$	N = number of routers (60 in all reported experiments)
Action space	$2N + 1$ discrete actions	N move actions, N send actions, 1 stay action
Trunk	$\text{Linear}(\text{state_dim}, 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128, 128) \rightarrow \text{LayerNorm}(128) \rightarrow \text{ReLU}$	Shared feature extractor
Value stream	$\text{NoisyLinear}(128, 128) \rightarrow \text{ReLU} \rightarrow \text{NoisyLinear}(128, 51)$	Distributional value logits
Advantage stream	$\text{NoisyLinear}(128, 128) \rightarrow \text{ReLU} \rightarrow \text{NoisyLinear}(128, \text{action_dim} \times 51)$	Distributional advantage logits
Distributional support	51 atoms over $[v_{\min}, v_{\max}] = [-100, 200]$	Categorical projection update
Noisy layers	Factorized NoisyLinear, $\text{std_init} = 0.5$	Exploration in value and advantage heads

A.3 Rainbow DQN Hyperparameters

Table A3 provides the complete training configuration for reproducibility.

Table A3 Complete Rainbow DQN hyperparameters for reproducibility.

Setting	Attacker policy	Scout policy	Implementation detail
Optimizer	Adam	Adam	Learning rate = 2×10^{-4}
Discount factor γ	0.99	0.97	Applied in n -step return construction
Replay capacity	50,000	50,000	FIFO replacement
Batch size	256	256	PER sampling each gradient step
PER α	0.6	0.6	Priority exponent
PER β (initial)	0.4	0.4	Importance-sampling correction
PER β increment/episode	0.001	0.004	Annealed by <code>next_episode()</code>
Priority ϵ	10^{-6}	10^{-6}	Added to absolute per-sample loss
Priority cap	1e6	none	Attacker priority clipping for stability
n -step horizon	up to 6	up to 6	Flush when $n \geq 6$ or terminal
Distributional atoms	51	51	Support $[-100, 200]$
Target sync cadence	200 updates; 500 after replay > 10,000	200 updates; 500 after replay > 10,000	Soft update every cadence trigger
Soft update rate τ	0.005	0.005	Applied to target parameters
Gradient handling	NaN-to-zero + clip-norm 10.0	NaN-to-zero + clip-norm 10.0	Before optimizer step
State normalization	StandardScaler	StandardScaler	Fitted after 1,500 buffered states
Exploration schedule (training)	same	same	Episodes < 10: $\epsilon = 1.0$; 10–509: $\epsilon = 0.05$; ≥ 510 : $\epsilon = 0.01$
Policy update frequency	every 10 ticks	every 4 ticks	Triggered in NetLogo training loop
Checkpointing cadence	every 50 episodes + final save	every 50 episodes + final save	Main checkpoint plus timestamped backup
Senior’s prior-study training regime	up to 4 agents/-side	up to 4 agents/-side	Regime under which the <i>frozen senior</i> was trained in the prior study; the junior policies of this study were trained at the fixed 5 attackers / 3 scouts formation (Section 3.1)
Senior’s prior-study role splits	sampled within 4-agent cap	sampled within 4-agent cap	Applies to the senior only; representative split 3 attackers + 1 scout, total capped at 4 per side

A.4 Reward Coefficients and Conditions

Table A4 lists the complete reward specification used in training for attacker and scout policies, including coefficient values, event conditions, and terminal bonuses. These values are the exact implementation settings used in the reported experiments.

Table A4 Complete reward function coefficients and trigger conditions used during RL training.

Term	Coefficient/value	Condition
Tick cost (attackers and scouts)	-0.01 per tick	Applied every tick to all attacker-side agents
Idle streak penalty	$-0.5 \times \text{stay_streak}$	Stay action, both attacker and scout policies
Repeat move penalty	$-1.5 \times \text{repeat_count}$	If same move target appears at least twice in recent window
Backtrack penalty	-2	Immediate A→B→A move pattern
Move-heavy window penalty	$-0.2 \times (\text{moves}_6 - 3)$	If more than 3 moves in last 6 actions
Inward move bonus	+2	Move to a router closer to center (both roles)
Oscillation penalty	-4	Revisiting same router within 3 ticks (both roles)
Move action cost	-0.2	Charged on move execution (both roles)
Scout send: unknown-state bonus	+3	Scout sends to router marked "unchecked"
Scout send: duplicate send penalty	$-2 \times \text{sent_count}$	Multiple scout sends to same target in current tick
Scout send: first-send bonus	+2	First send to target in current tick
Scout send: center-neighbor bonus	+3	Target adjacent to center router
Scout send: recency penalty	-2	Same target sent within last 3 ticks
Scout freshness shaping	$+ \left(1 - \frac{\text{last_rep}}{\tau_f}\right) - \left(2 + \frac{\text{overshoot}}{\tau_f}\right)$	or Uses freshness threshold τ_f
Scout infection-neighbor bonus	+6	Target adjacent to recently infected router and currently unchecked
Attacker send: stale-awareness penalty	-2	Last scout report older than freshness threshold
Attacker send: fresh-awareness bonus	$+ \left(1 - \frac{\text{last_rep}}{\tau_f}\right)$	If report age within freshness threshold
Attacker send: vulnerable target bonus	+4	Target marked vulnerable in shared awareness
Attacker send: clean target penalty	-2	Target marked clean in shared awareness
Attacker send: center-neighbor bonus	+2	Target adjacent to center router
Attacker send: recent-infection-neighbor bonus	+8	Target adjacent to recent infection and marked vulnerable
Failed attack penalty (invalid target)	-5 or -3	-5 if offline/already infected; -3 if not vulnerable
Successful infection reward	+25	Router successfully infected
Center-compromise bonus	additional +25	If infected router is designated center
Scout assist bonus	+scout_infection_assist_bonus (default 2)	Scout reported same router shortly before infection
Scout terminal coverage bonus	$+30 \times \text{coverage_fraction}$	Added at episode end for scout packet transitions
Team terminal bonus (attackers)	+20 win / -20 loss	Terminal synthetic transition
Team terminal bonus (scouts)	+10 win / -10 loss	Terminal synthetic transition